

Limits of Representability

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Representability and semi-representability

Given a theory T in a signature Σ extending Str , and an n -ary relation R on strings:

Definition

R is **representable in** T iff there is a definitional extension T^+ of T with a new n -ary predicate F such that:

- ▶ Whenever $Rs_1 \dots s_n$, $T^+ \models F(\langle s_1 \rangle, \dots, \langle s_n \rangle)$
- ▶ Whenever it's not the case that $Rs_1 \dots s_n$, $T^+ \models \neg F(\langle s_1 \rangle, \dots, \langle s_n \rangle)$

Definition

R is **semi-representable in** T iff there is a definitional extension T^+ of T with a new n -ary predicate F such that:

- ▶ Whenever $Rs_1 \dots s_n$, $T^+ \models F(\langle s_1 \rangle, \dots, \langle s_n \rangle)$
- ▶ Whenever it's not the case that $Rs_1 \dots s_n$, $T^+ \not\models F(\langle s_1 \rangle, \dots, \langle s_n \rangle)$

Where T is a theory in a signature extending Str , and g is partial function from n -tuples of strings to strings:

Definition

g is **capturable in** T iff there is a definitional extension T^+ of T with a new n -ary function symbol f such that:

- Whenever $t = g(s_1, \dots, s_n)$, $T^+ \models \langle t \rangle = f(\langle s_1 \rangle, \dots, \langle s_n \rangle)$

Equivalently: if there is an $n + 1$ -formula P with free variables $v_1, \dots, v_{n+1}$ such that

- (i) $T \models \forall v_1 \dots \forall v_n \exists! v_{n+1} P$, and
- (ii) Whenever $t = g(s_1, \dots, s_n)$, $T \models P[\langle s_1 \rangle / v_1, \dots, \langle s_n \rangle / v_n, \langle t \rangle / v_{n+1}]$.

Note: In the book, this is called ‘representability’ too; but this is confusing given that partial functions are relations.

A few simple observations about these concepts

(iv) if $T \subseteq T^+$, every relation representable in T is representable in T^+ . However, some relations semi-representable in T may not be semi-representable in T^+ .

Some observations about these concepts

(i) If T captures g and g' , it captures $g' \circ g$.

- Definitionally extend T with function symbols f and f' such that $T^+ \models f(\langle s \rangle) = \langle g(s) \rangle$ and $T^+ \models f'(\langle s \rangle) = \langle g'(s) \rangle$ for all s . Then further definitionally extend with the definition

$$\forall x \forall y (y = f''(x) \leftrightarrow y = f'(f(x)))$$

(ii) If T captures a function f and (semi-)represents X , it (semi-)represents $\{y \mid fy \in X\}$ (the preimage of X under f —sometimes written $f^*(X)$).

- Definitionally extend T with a function symbol f such that $T^+ \models f(\langle s \rangle) = \langle g(s) \rangle$ and a predicate F such that $T^+ \models F(\langle s \rangle)$ whenever $s \in X$ and $T^+ \models \neg F(\langle s \rangle)$ ($T^+ \not\models F(\langle s \rangle)$) otherwise. Then further definitionally extend with the definition

$$\forall x (Gx \leftrightarrow F(f(x)))$$

Tarski's non-representability theorem

Consequences of Cantor's Theorem

When T is a theory in a signature extending Str , and P is a formula with free variables $v_1, \dots, v_n$ (in alphabetical order), and $s_1, \dots, s_n$ are strings, say that P is **T -provable of $s_1, \dots, s_n$** iff $T \models P[\langle s_1 \rangle / v_1, \dots, \langle s_n \rangle / v_n]$.

Each of the countably many 1-formulae semi-represents at most one set of strings, and there are uncountably many sets of strings, so by Cantor's theorem, some sets of strings aren't semi-representable in T (and thus aren't representable in T if T is consistent).

And we can give an example! Consider any set Y that contains all 1-formula that are *not T -provable of themselves*, and no other 1-formulae. Y isn't semi-representable in T , since if 1-formula $NPOS(x)$ represented it, we would have both

- ▶ $NPOS(x) \in Y$ iff $T \not\models NPOS(\langle NPOS(x) \rangle)$ (by the definition of Y).
- ▶ $T \models NPOS(\langle NPOS(x) \rangle)$ iff $NPOS(x) \in Y$ (since $NPOS(x)$ semi-represents Y).

Note that if T is consistent, it follows that Y is not representable in T .

More non-representable and non-semi-representable sets and relations

- ▶ Consider now the set of all 1-formulae that *are* T -provable of themselves. If T is consistent, it can't be representable in T , since if it were, its complement would be too, which we just ruled out. (However it could still be *semi*-representable.)
- ▶ The relation P is T -provable of Q also can't be representable in T if T is consistent (though it could be semi-representable). For if it were represented by a 2-formula $\text{ProvOf}_T(x, y)$, the 1-formula $\neg \text{ProvOf}(x, x)$ would represent the set of all 1-formulae that are T -provable of themselves.
- ▶ The relation P is *not* T -provable of Q can't even be semi-representable in T . For if it were semi-represented by $\text{NotProvOf}_T(x, y)$, $\text{NotProvOf}_T(x, x)$ would semi-represent the set of 1-formulae not T -provable of themselves.

Let an x -formula be a formula in which the only free variable is x .

Definition

For any signature Σ , Σ 's *self-application* is the function that maps each x -formula P to the sentence $P[\langle P \rangle/x]$.

We are going to be interested in theories that *capture self-application*, which means they have a definitional expansion with a function symbol `SelfApply` such that for every x -formula P ,

$$T \models \langle P[\langle P \rangle/x] \rangle = \text{SelfApply}(\langle P \rangle)$$

Labelling, substitution, and self-application

Note that any theory that can capture the *labelling* function $\langle \cdot \rangle$ and the *substitution* function that takes a formula P , a variable v , and a term t and yields $P[t/v]$ can also capture self-application. For consider a definitional extension of T with an 1-ary function symbol Label and 3-ary function symbol Subst , and now introduce a further definition:

$$\forall y \forall z (z = \text{SelfApply}(y) \leftrightarrow z = \text{Subst}(y, \langle x \rangle, \text{Label}(y)))$$

For each x -formula P , we'll have $T \models \langle \langle P \rangle \rangle = \text{Label}(\langle P \rangle)$ and $T \models \langle P[\langle P \rangle/x] \rangle = \text{Subst}(\langle P \rangle, \langle x \rangle, \langle \langle P \rangle \rangle)$, hence $T \models \langle P[\langle P \rangle/x] \rangle = \text{SelfApply}(\langle P \rangle)$.

Later, we'll prove that *Min* *does* capture labelling and substitution, and hence captures self-application. (It follows that the same is true for all theories extending *Min*.)

Tarski's non-representability theorem

Tarski's Non-Self-Representability Theorem

No consistent theory that captures self-application represents itself.

Proof: suppose for contradiction that T is consistent, function symbol `SelfApply` captures self-application in T , and predicate Prov_T represents T in T .

Now consider the 1-formula $\neg \text{Prov}_T(\text{SelfApply}(x))$.

Given our assumptions, it would have to represent a set containing all 1-formulae whose self-applications are not in T (i.e., which are not T -provable of themselves) and no other 1-formulae. But we've already shown that no such set is representable (given that T is consistent).

Non-Semi-Representability Theorem

No theory T captures self-application and semi-represents any set of strings that doesn't contain any member of T and contains all sentences in T 's signature that aren't in T .

Suppose X is such a set, semi-represented in (a definitional extension of) T by predicate F . Then the 1-formula $F(\text{SelfApply}(x))$ would semi-represent a set containing all 1-formulae whose self-application is in X , i.e. which are not T -provable of themselves, and no other 1-formulae.

Non-Self-Semi-Representability Theorem

No consistent theory that captures self-application and semi-represents itself is negation-complete.

Suppose T is negation complete and consistent, and has a function symbol SelfApply and predicate Prov_T that respectively capture self-application and semi-represent T in T . Then the 1-formula $\text{Prov}_T(" \neg " \oplus \text{SelfApply}(x))$ would semi-represent the set of all 1-formulae such that the negation of their self-application is a member of T . Since T is consistent, this set doesn't include any 1-formulae whose self-application is in T , and since (since T is negation-complete and closed under double negation elimination) it includes every 1-formula whose self-application isn't in T . But this is impossible.

The diagonal lemma

The Diagonal Lemma

A different and useful route to the same result goes via the *diagonal lemma*, also known as the *fixed point lemma*. Define:

Definition

Sentence G is a **fixed point** of x -formula $H(x)$ in T iff $T \models G \leftrightarrow H(\langle G \rangle)$.

- It's traditional (though potentially misleading) to think of a sentence G that is the fixed point of $H(x)$ in T as “saying of itself that it is H ”—as if it were the sentence ‘I am H ’ or ‘This very sentence is H ’.

Generalizing an idea that we in effect already employed in the Unrepresentability Theorem, we can prove

The Diagonal Lemma

If T captures self-application, then every 1-formula has a fixed point in T .

Proof of the Diagonal Lemma

Proof: Definitionally extend T with a function symbol SelfApply that captures self-application, and for any 1-formula $H(x)$, let G_H be the sentence

$$H(\text{SelfApply}(\langle H(\text{SelfApply}(x)) \rangle))$$

i.e., the self-application of the 1-formula $H(\text{SelfApply}(x))$. Call this the **diagonalization** of $H(x)$.

Since SelfApply captures self-application in T ,

$$T \models \text{SelfApply}(\langle H(\text{SelfApply}(x)) \rangle) = \langle G \rangle$$

Since T is closed under logical consequence, this implies

$$T \models H(\text{SelfApply}(\langle H(\text{SelfApply}(x)) \rangle)) \leftrightarrow H(\langle G_H \rangle)$$

i.e.

$$T \models G_H \leftrightarrow H(\langle G_H \rangle)$$

From the Diagonal Lemma to the Unrepresentability Theorem

Suppose for contradiction that T captures self-application (with a function symbol SelfApply) and also represented T itself (with a predicate Prov_T). By the Diagonal Lemma, the formula $\neg \text{Prov}_T(x)$ has a fixed point in T . That is: a sentence G_T such that

$$T \models G_T \leftrightarrow \neg \text{Prov}_T \langle G_T \rangle$$

Since T is a theory, it follows that

$$T \models G_T \text{ iff } T \models \neg \text{Prov}_T(\langle G_T \rangle)$$

Suppose $T \models G_T$. Then $T \models \text{Prov}_T(\langle G_T \rangle)$ (since Prov_T represents T in T), and also $T \models \neg \text{Prov}_T(\langle G_T \rangle)$ (by the above biconditional), so T is inconsistent, contradicting our assumption.

Suppose on the other hand that $T \not\models G_T$. Then, $T \models \neg \text{Prov}_T \langle G_T \rangle$ (since Prov_T represents T in T), so $T \models G_T$ by the above biconditional: contradiction.

What Min represents

The next order of business is to show that Min can represent the labelling and substitution functions (and hence also self-application).

We will also show that Min represents the set of *proofs* (in our proof system $\vdash$). And as a consequence of this: for any set Ax that's representable in Min, Min represents the relation *p is a proof whose final sequent has Q on the right and only members of Ax on the left.*

A version of Gödel's theorem

Suppose that a theory T that extends Min is axiomatised by a set Ax that's representable in Min (and hence also in T). (For example, Ax could be any finite set). Let $\text{Proof}_{Ax}(x, y)$ be a predicate that represents the relation *p is a proof of Q from Ax*. Then the predicate Prov_T defined by $\exists x \text{Proof}_{Ax}(x, y)$ defines T , and also *semi-represents* T in Min .

This means that we have $T \models \text{Prov}_T(\langle P \rangle)$ for all $P \in T$. So in particular, if $T \models G_T$ (where G_T is the diagonalization of $\neg \text{Prov}_T(x)$), then T is inconsistent. So G_T is not in T , so there's no proof of it from Ax . It follows from this that it is true in $\mathbb{S}$. So we have a variant of Gödel's theorem: no consistent finitely (or more generally, Min -representably) axiomatizable theory that extends Min contains every sentence true in $\mathbb{S}$.